

# SIVANESAN L

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## CAREER OBJECTIVE

Motivated Electronics and Communication Engineering student with experience in Python, Flask, PostgreSQL, cloud-based systems, and IoT projects. Seeking a Software Developer role to build scalable applications and solve real-world problems.

## EDUCATION

B.E. Electronics and Communication Engineering — Anna University | 2022–2026

Higher Secondary (12th) — State Board | 2022 | 79%

SSLC (10th) — State Board | 2020 | 86%

## TECHNICAL SKILLS

Programming Languages: Python, C, Java (Basics), JavaScript (Basics)

Frontend: HTML, CSS

Database: PostgreSQL

Core Concepts: OOP, Problem Solving, Data Structures (Python Basics)

Frameworks & Libraries: Flask, FastAPI, PyQt5, SQLAlchemy

Tools & Tech: Git, GitHub, Linux basics, ESP32, Blynk, IoT, REST APIs, JWT Authentication, OAuth2

## INTERNSHIP EXPERIENCE

Embedded Systems & IoT Intern | Techvolt Software Pvt. Ltd

June 2025 – July 2025

- Developed a Smart Intrusion Detection System using IR sensor, ultrasonic sensor, and servo motor
- Integrated multiple sensors with ESP32 for accurate intrusion monitoring

## PROJECTS

### Infrastructure-Less Smart Surveillance System | Final Year Project

- Designed a cloud-based surveillance system eliminating the need for traditional NVR/DVR hardware
- Streamed IP camera frames to a Flask server hosted on Google Colab using ngrok tunneling
- Implemented real-time facial recognition using InsightFace for known and unknown face detection
- Integrated Twilio API for SMS and voice call alerts and Telegram Bot API for unknown face image notifications
- Used Google Drive structured storage for face datasets and event logs

Tech: Python, Flask, InsightFace, OpenCV, Google Colab, Ngrok, Twilio API, Telegram Bot API, Cloud Storage

### Expense Tracker REST API

- Developed a secure Expense Tracker REST API using FastAPI and PostgreSQL
- Implemented JWT-based authentication and OAuth2 authorization for secure user access
- Designed one-to-many relationships between users and expenses using SQLAlchemy ORM
- Built asynchronous CRUD operations for efficient expense management
- Secured API endpoints to ensure users can access only their own expense records

Tech: FastAPI, PostgreSQL, SQLAlchemy, JWT, OAuth2, Python

### Student Data Management System | Flask Web Application

- Developed a web-based Student Data Management System using Flask and PostgreSQL
- Implemented CRUD operations for efficient student record management
- Designed database integration for storing, retrieving, updating, and deleting student information
- Built responsive web forms for managing student records through a user-friendly interface
- Followed MVC architecture principles and backend development best practices

Tech: Flask, Python, PostgreSQL, HTML, CSS

## ACHIEVEMENTS

- Completed Industry 4.0 IoT project under Naan Mudhalvan and internship training in Embedded Systems & IoT Design

## SOFT SKILLS

Problem solving, analytical thinking, teamwork, communication, adaptability, self-learning

## LANGUAGES

Tamil (Native), English (Professional Working)